

ENGE702 Engineering Mathematics III

Tutorial 5

Note: the first two questions are revision on basic matrix operations. Skip these if you are comfortable with matrix basics.

1. Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \\ -4 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 3 & -2 \end{pmatrix}$, $C = \begin{pmatrix} -4 & 2 & 1 \\ 2 & -1 & 5 \end{pmatrix}$, $D = \begin{pmatrix} 6 & 1 & 5 \\ 2 & 2 & -4 \\ 1 & 3 & -2 \end{pmatrix}$.

Calculate, if possible: (hint: consult EM1 notes)

(a) AB $\begin{pmatrix} -1 & 2 \\ 9 & -6 \\ -1 & -2 \end{pmatrix}$ (d) DA $\begin{pmatrix} -8 & 2 \\ 20 & 0 \\ 10 & 6 \end{pmatrix}$ (g) $A + C$ not defined
 (b) AC $\begin{pmatrix} -10 & 5 & -3 \\ 6 & -3 & 15 \\ 18 & -9 & 1 \end{pmatrix}$ (e) $AC + B$ not defined (h) $A^T + C$ $\begin{pmatrix} -2 & 2 & -3 \\ 1 & 2 & 6 \end{pmatrix}$
 (c) AD not defined (f) $AB + B$ not defined (i) $A + C^T$ $\begin{pmatrix} -2 & 1 \\ 2 & 2 \\ -3 & 6 \end{pmatrix}$

2. Find the inverses of the following matrices, if they exist: (hint: consult EM1 notes)

(a) $\begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}$ $\begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & -\frac{1}{2} \end{pmatrix}$ (b) $\begin{pmatrix} -1 & 3 \\ 2 & -6 \end{pmatrix}$ does not exist (c) $\begin{pmatrix} 6 & 0 \\ 2 & 1 \end{pmatrix}$ $\begin{pmatrix} \frac{1}{6} & 0 \\ -\frac{1}{3} & 1 \end{pmatrix}$

3. Find eigenvalues and eigenvectors for the following matrices:

(hint: follow the process from the second half of lecture 9, where we used $\begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix}$)

(a) $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ $\lambda_1 = 1$ $\lambda_2 = 3$ $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (b) $\begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$ $\lambda_1 = -1$ $\lambda_2 = -2$ $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ (c) $\begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix}$ $\lambda_1 = 3$ $\lambda_2 = -2$ $v_1 = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

4. Find the general solution to the following systems of differential equations:

(hint: follow the process from lecture 10, where we used $\begin{matrix} y_1' = 7y_1 - y_2 \\ y_2' = 3y_1 + 3y_2 \end{matrix}$)

(a) $\begin{matrix} y_1' = 3y_1 + 3y_2 \\ y_2' = 5y_1 + y_2 \end{matrix}$ $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{6t} + c_2 \begin{bmatrix} 1 \\ \frac{5}{3} \end{bmatrix} e^{-2t}$ (b) $\begin{matrix} y_1' = -3y_1 + y_2 \\ y_2' = y_1 - 3y_2 \end{matrix}$ $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-4t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-2t}$

5. Solve the following initial value problems:

(hint: follow the process from lecture 10 that immediately followed the example we mentioned in question 2 above)

(a) $\begin{matrix} y_1' = -2y_1 + y_2 \\ y_2' = 9y_1 - 2y_2 \end{matrix}$ $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix} e^t + \begin{bmatrix} -\frac{1}{2} \\ \frac{3}{2} \end{bmatrix} e^{-5t}$ (b) $\begin{matrix} y_1' = y_1 + 3y_2 \\ y_2' = 0y_1 + 2y_2 \end{matrix}$ $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} e^t + \begin{bmatrix} 3 \\ 1 \end{bmatrix} e^{2t}$
 $y_1(0) = 0, \quad y_2(0) = 3$ $y_1(0) = 2, \quad y_2(0) = 1$

6. Find the general solution to the following differential equations by converting to a system of two first order differential equations and solving:

(hint: follow the process from the end of lecture 10, where we used $y'' + 5y' + 4y = 0$)

(a) $y'' - 7y' + 10y = 0$ $y = c_1 e^{2t} + c_2 e^{5t}$
 (b) $y'' + 6y' + 8y = 0$ $y = c_1 e^{-4t} + c_2 e^{-2t}$
 (c) $y'' + 5y' + 6y = 0$ $y = c_1 e^{-2t} + c_2 e^{-3t}$

7. Now for *another* mixing problem. We have two tanks, each containing 100l of fluid. Tank 1 initially contains pure water and tank 2 initially contains water with 50kg of salt dissolved. The tanks are stirred constantly so their solutions have uniform concentration. There is a pipe which takes 15 l/min from tank 1 to tank 2, and another pipe which takes 5 l/min from tank 2 to tank 1. Also, pure water is being pumped into tank 1 at a rate of 10 l/min, and solution is being removed at a rate of 10 l/min from tank 2.

Find the concentration of salt in tank 1 as a function of time.

(hint: follow the mixing problem we did in lecture 10, draw a picture and work out what is coming in and what is going out)

$$y_1 = \frac{25}{\sqrt{3}} e^{-0.0634t} - \frac{25}{\sqrt{3}} e^{-0.236t}$$

Computer Part

Even though tutorials are not run in computer labs, during weeks 1 - 6 of the semester, you should work through the section 'Solving Ordinary Differential Equations With MATLAB' (under 'Computational Mathematics') in the self paced courses on the mathworks website, as 'homework'.

If you have never used matlab before, it would be a good idea to work through the section 'MATLAB Onramp' first. Details of how to access these are as follows:

AUT has a licence with the matlab creators that allows you to access an array of online training material, and also download the latest version of matlab to your personal computer (if you wish - you don't need to do this for this course if you don't want to, you can access matlab on the computers at AUT).

To access these resources you will need to create an account with mathworks (the company that produces matlab).

You can do this by visiting [mathworks.com](https://www.mathworks.com) (or google 'matlab'). You can create an account by clicking the 'login' icon on the top right. **It is important you use your AUT email address to create the account, or you will not be able to access the material.**

From your account page, click 'Self-Paced Courses'.